

Part A: Numeric Expressions*Evaluate each expression without the use of a calculator.*

1. $5 + 4$

15. $6 \cdot (-8)$

2. $-7 - 5$

16. $5 \cdot (-6)$

3. $6 - 4$

17. $5^3 - (-11) \cdot 5$

4. $12 - (-1)$

18. $-11 - 7 \cdot ((-4) + (-11))$

5. $11 + (-9)$

19. $(-7)^3 - 15 \cdot 2$

6. $-9 + 8$

20. $13 - 8 \cdot (3 + 13)$

7. $9 - (-11)$

21. $-11 - 6 \cdot (3 + (-11))$

8. $10 - 9$

22. $(7 + 17)^2 \div 4$

9. $8 \cdot (-5)$

23. $-8 - 12 \cdot (7 + (-8))$

10. $36 \div (-6)$

24. $-9 - 6 \cdot (9 + (-9))$

11. $-8 \div (-2)$

25. $4^2 - 7 \cdot 9$

12. $6 \cdot (-2)$

26. $(4 + 18)^2 \div (-11)$

13. $-42 \div 6$

27. $14 - 9 \cdot ((-3) + 14)$

14. $5 \cdot (-7)$

28. $-13 - 7 \cdot ((-7) + (-13))$

29. $(7 + 17)^2 \div 1$

34. $(4 + (-18))^2 \div (-7)$

30. $(5 + (-3))^2 \div (-2)$

35. $5^2 - 12 \cdot 6$

31. $(9 + (-10))^2 \div 1$

36. $2^2 - (-7) \cdot 6$

32. $(-4 + 18)^2 \div (-14)$

37. $3 - 11 \cdot ((-7) + 3)$

33. $(-6 + (-6))^2 \div 6$

38. $(-8)^2 - 5 \cdot 3$

Part B: Evaluating Expressions

Evaluate each using the values given.

1. $-j - j + 4j - y^2 + 6, \quad j = 6, \quad y = -3$

7. $c + 4c - y^2 + y, \quad c = -1, \quad y = -5$

2. $4m^2 - 4m - 2q - q + 7, \quad m = -2, \quad q = -1$

8. $4m^2 - 4m - 4p - 2p - 8, \quad m = 2, \quad p = 7$

3. $-4n - 2n + 3n + 3y^2 + 3, \quad n = 2, \quad y = 3$

9. $-2c - c - 3y^2, \quad c = 6, \quad y = 5$

4. $3j^2 - j - j - y - 1, \quad j = 7, \quad y = 4$

10. $b + 3b - 3n^2 - 2n - 4, \quad b = -4, \quad n = 6$

5. $-2k^2 + 2k - x + x - 1, \quad k = -6, \quad x = 5$

11. $-3j - 2j + 3j - x^2 - 4, \quad j = 2, \quad x = -2$

6. $-4j - 3j + 2m^2 - 6, \quad j = 4, \quad m = 4$

12. $-4p + 3p - 4x^2 + 8, \quad p = -7, \quad x = 6$

Part C: Simplifying Expressions

Simplify each expression.

1. $-7x + 5 - 3x + 6$

3. $x + 5 - 6x + 7$

2. $8x + 2 + 5x + 3$

4. $8x - 1 + 5x - 3$

19. $3x - 2 + 12x + 7$

5. $8x - 6 + 4x + 1$

20. $4x - 3 + 5x + 8$

6. $x + 2 + 5x + 3$

21. $16x - 14y + 7 - 8x + 2y$

7. $4x + 3y - 10 + 11x - 9y$

22. $7x - 11y + 8 + 10x - 14y$

8. $8x + 11y + 2 + 5x + 10y$

23. $4x - 17y + 2 + 9x + 8y$

9. $-6x + 5y - 4 + 7x + 4y$

24. $18x + 20y + 4 + 3x + 9y$

10. $9x + 12y + 7 - 12x - 2y$

25. $10x - 2y - 5 + 3x + 20y$

11. $12x + 10y + 3 - x + 8y$

26. $2x + 9x - 11x^2 + 3x^2$

12. $3x - 11y - 5 + 2x + 10y$

27. $5x + 6x^2 + 2x - 12x^2$

13. $7x + 2y + 8 - 5x - 8y$

28. $-8x^2 + 4x + x - 7x^2$

14. $8x + 11y + 1 + 5x + 9y$

29. $3x + 6x - 10x^2 + 4x^2$

15. $10x + 4y + 3 + 11x + 5y$

30. $10x - 9x^2 + 2x^2 + 5x$

16. $-8x + 2y + 5 - 6x + 8y$

31. $-10x + 11x^2 + 12x^2 + 5x$

17. $-x - 8y + 10 + 2x + 8y$

32. $5x + 11x^2 + x + 8x^2$

18. $-7x + 4 + x + 6$

33. $10x^2 + 9x + 7x^2 + 2x$

34. $4x + 6x + 7x^2 + 2x^2$

43. $19x^2 + 18x^2 + 12x^3 + 15x^3 - 19x - 6x$

35. $12x^2 + 3x^2 + 10x + 7x$

44. $18x + 2x^3 + 13x + 15x^3 + 12x^2 + 5x^2$

36. $10x + 2x^2 + x^2 + 8x$

45. $12x^2 - 20x + 3x^3 + 16x^3 + 8x + 17x^2$

37. $-x + 8x^2 - 12x^2 + 5x$

46. $-10x^3 - 15x^2 + 8x^2 + 19x + 12x^3 - 17x$

38. $11x^2 + 12x + 2x - 6x^2$

47. $-10x^3 + 2x - 13x^3 + 8x + 9x^2 - 2x^2$

39. $18x^3 + 9x^2 - 20x^3 + 8x - 18x + 7x^2$

48. $-18x^3 + 4x^2 + 19x + 12x^3 + 7x^2 + 14x$

40. $-14x^2 - x^3 + 7x + 2x^2 + 15x + 19x^3$

49. $11x^2 + 13x^2 + 19x^3 - 3x^3 + 5x + 10x$

41. $18x - 12x^2 + 16x^3 - 20x + 8x^2 + 10x^3$

50. $4x^3 - 11x^3 - 17x^2 + 7x^2 - 12x + 3x$

42. $4x^2 + 10x + 11x^3 + 5x^2 - 14x - 4x^3$

Part D: Properties of Addition and Multiplication

Select the property that is displayed in the given equation.

1. $y + 12 = 12 + y$

- A) Commutative Property of Addition
- B) Inverse Property of Addition
- C) Associative Property of Addition
- D) Identity Property of Addition

2. $y \cdot b = b \cdot y$

- A) Inverse Property of Multiplication
- B) Identity Property of Multiplication
- C) Commutative Property of Multiplication
- D) Associative Property of Multiplication

3. $11 + (-13) = -13 + 11$

- A) Commutative Property of Addition
 - B) Inverse Property of Addition
 - C) Associative Property of Addition
 - D) Identity Property of Addition
4. $-11 + 8 = 8 + (-11)$
- A) Identity Property of Addition
 - B) Associative Property of Addition
 - C) Commutative Property of Addition
 - D) Inverse Property of Addition
5. $z \cdot 3 = 3 \cdot z$
- A) Inverse Property of Multiplication
 - B) Identity Property of Multiplication
 - C) Associative Property of Multiplication
 - D) Commutative Property of Multiplication
6. $3 + z = z + 3$
- A) Commutative Property of Addition
 - B) Identity Property of Addition
 - C) Associative Property of Addition
 - D) Inverse Property of Addition
7. $(1 + 0) + 13 = 1 + (0 + 13)$
- A) Inverse Property of Addition
 - B) Commutative Property of Addition
 - C) Identity Property of Addition
 - D) Associative Property of Addition
8. $(0 + 15) + 13 = 0 + (15 + 13)$
- A) Inverse Property of Addition
 - B) Associative Property of Addition
 - C) Commutative Property of Addition
 - D) Identity Property of Addition
9. $(b \cdot a) \cdot y = b \cdot (a \cdot y)$
- A) Identity Property of Multiplication
 - B) Commutative Property of Multiplication
 - C) Associative Property of Multiplication
 - D) Inverse Property of Multiplication
10. $(-9 + 7) + 10 = -9 + (7 + 10)$
- A) Inverse Property of Addition
 - B) Commutative Property of Addition
 - C) Identity Property of Addition
 - D) Associative Property of Addition
11. $(-5 + a) + (-9) = -5 + (a + (-9))$

- A) Identity Property of Addition
 - B) Inverse Property of Addition
 - C) Associative Property of Addition
 - D) Commutative Property of Addition
12. $(b + 6) + (-1) = b + (6 + (-1))$
- A) Inverse Property of Addition
 - B) Identity Property of Addition
 - C) Associative Property of Addition
 - D) Commutative Property of Addition
13. $-13 \cdot 1 = -13$
- A) Commutative Property of Multiplication
 - B) Inverse Property of Multiplication
 - C) Associative Property of Multiplication
 - D) Identity Property of Multiplication
14. $5 \cdot 1 = 5$
- A) Commutative Property of Multiplication
 - B) Associative Property of Multiplication
 - C) Identity Property of Multiplication
 - D) Inverse Property of Multiplication
15. $-4 + 0 = -4$
- A) Inverse Property of Addition
 - B) Identity Property of Addition
 - C) Associative Property of Addition
 - D) Commutative Property of Addition
16. $6 + 0 = 6$
- A) Associative Property of Addition
 - B) Inverse Property of Addition
 - C) Commutative Property of Addition
 - D) Identity Property of Addition
17. $0 + (-7) = -7$
- A) Identity Property of Addition
 - B) Associative Property of Addition
 - C) Inverse Property of Addition
 - D) Commutative Property of Addition
18. $0 + (-7) = -7$
- A) Associative Property of Addition
 - B) Identity Property of Addition
 - C) Inverse Property of Addition
 - D) Commutative Property of Addition
19. $z \cdot \frac{1}{z} = 1$

- A) Inverse Property of Multiplication
- B) Commutative Property of Multiplication
- C) Associative Property of Multiplication
- D) Identity Property of Multiplication

20. $5 + (-5) = 0$

- A) Inverse Property of Addition
- B) Associative Property of Addition
- C) Identity Property of Addition
- D) Commutative Property of Addition

21. $8 + (-8) = 0$

- A) Associative Property of Addition
- B) Commutative Property of Addition
- C) Identity Property of Addition
- D) Inverse Property of Addition

22. $-6 \cdot \frac{1}{-6} = 1$

- A) Commutative Property of Multiplication
- B) Associative Property of Multiplication
- C) Inverse Property of Multiplication
- D) Identity Property of Multiplication

23. $10 \cdot \frac{1}{10} = 1$

- A) Associative Property of Multiplication
- B) Inverse Property of Multiplication
- C) Identity Property of Multiplication
- D) Commutative Property of Multiplication

24. $z \cdot \frac{1}{z} = 1$

- A) Commutative Property of Multiplication
- B) Identity Property of Multiplication
- C) Inverse Property of Multiplication
- D) Associative Property of Multiplication

25. $-3 + a = a + (-3)$

- A) Associative Property of Addition
- B) Identity Property of Addition
- C) Commutative Property of Addition
- D) Inverse Property of Addition

26. $(-6 + 2) + y = -6 + (2 + y)$

- A) Identity Property of Addition
- B) Inverse Property of Addition
- C) Associative Property of Addition
- D) Commutative Property of Addition

27. $a \cdot (-9) = -9 \cdot a$
A) Identity Property of Multiplication
B) Commutative Property of Multiplication
C) Associative Property of Multiplication
D) Inverse Property of Multiplication
28. $1 \cdot 3 = 3$
A) Inverse Property of Multiplication
B) Associative Property of Multiplication
C) Commutative Property of Multiplication
D) Identity Property of Multiplication
29. $-9 \cdot \frac{1}{-9} = 1$
A) Inverse Property of Multiplication
B) Associative Property of Multiplication
C) Identity Property of Multiplication
D) Commutative Property of Multiplication
30. $(-9 \cdot 12) \cdot 13 = -9 \cdot (12 \cdot 13)$
A) Associative Property of Multiplication
B) Identity Property of Multiplication
C) Commutative Property of Multiplication
D) Inverse Property of Multiplication

Part E: Solving Equations

Solve each equation.

- | | |
|-----------------|-----------------|
| 1. $5x = -30$ | 6. $8x = -16$ |
| 2. $9x = 18$ | 7. $x + 3 = 7$ |
| 3. $x + 4 = 10$ | 8. $x + 9 = 12$ |
| 4. $7x = 28$ | 9. $x - 9 = -5$ |
| 5. $5x = 40$ | 10. $7x = -56$ |

11. $7x + 8 = 50$

23. $-4x + 3x = -3$

12. $8x + 1 = 49$

24. $33 = 8x + 7 + 2$

13. $2x - 4 = -6$

25. $57 = -5x + 2 - 5$

14. $6x - 1 = -49$

26. $130 = -12x + 5 - 7$

15. $5x - 9 = 26$

27. $12x + 9 + 12 = 18$

16. $10x + 4 = -\frac{98}{3}$

28. $\frac{14}{3} = 11x + 3x$

17. $4x - 7 = 4$

29. $8x + 3x = 88$

18. $12x - 10 = 86$

30. $21 = -9x + 12x$

19. $12x + 7 = 91$

31. $3(7x - 5) - 3 = 24$

20. $10x - 2 = -\frac{86}{3}$

32. $3(6x - 3) - 8 = -89$

21. $29 = 8x + 6 + 7$

33. $6(2x - 5) - 7 = -85$

22. $22 = 7x - 1 - 5$

34. $3(3x + 4) + 9 = 30$

35. $9(9x + 7) + 9 = -495$

36. $3(2x + 7) + 5 = 38$

44. $-\frac{1}{4}x + 9 = \frac{15}{2}$

37. $7(6x + 1) + 6 = -155$

45. $\frac{x - 6}{4} + 4 = \frac{5}{4}$

38. $7(x + 9) + 11 = 158$

46. $\frac{8x + 62}{4} = \frac{18}{12}$

39. $6(3x - 12) + 1 = 73$

47. $\frac{x - 2}{4} = \frac{4x - 29}{2}$

40. $7(4x + 4) - 8 = -36$

48. $\frac{-3}{x} = \frac{1}{3}$

41. $-2x - 6 = -14$

42. $\frac{x}{4} - 7 = -\frac{29}{4}$

49. $\frac{8x - 51}{2} = \frac{5}{2}$

43. $\frac{x}{2} + 1 = 2$

50. $\frac{56}{x} = \frac{7}{6}$

Answer Key

Part A: Numeric Expressions

- | | | |
|---------|----------|---------|
| 1. 9 | 14. -35 | 27. -85 |
| 2. -12 | 15. -48 | 28. 127 |
| 3. 2 | 16. -30 | 29. 576 |
| 4. 13 | 17. 180 | 30. -2 |
| 5. 2 | 18. 94 | 31. 1 |
| 6. -1 | 19. -373 | 32. -14 |
| 7. 20 | 20. -115 | 33. 24 |
| 8. 1 | 21. 37 | 34. -28 |
| 9. -40 | 22. 144 | 35. -47 |
| 10. -6 | 23. 4 | 36. 46 |
| 11. 4 | 24. -9 | 37. 47 |
| 12. -12 | 25. -47 | 38. 49 |
| 13. -7 | 26. -44 | |

Part B: Evaluating Expressions

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|--------|--------|----------|
| 1. 9 | 5. -85 | 9. -93 |
| 2. 34 | 6. -2 | 10. -140 |
| 3. 24 | 7. -35 | 11. -12 |
| 4. 128 | 8. -42 | 12. -129 |

Part C: Simplifying Expressions

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|----------------------|---------------------|---------------------------|
| 1. $11 - 10x$ | 18. $10 - 6x$ | 35. $15x^2 + 17x$ |
| 2. $13x + 5$ | 19. $15x + 5$ | 36. $3x^2 + 18x$ |
| 3. $12 - 5x$ | 20. $9x + 5$ | 37. $-4x^2 + 4x$ |
| 4. $13x - 4$ | 21. $8x - 12y + 7$ | 38. $5x^2 + 14x$ |
| 5. $12x - 5$ | 22. $17x - 25y + 8$ | 39. $-2x^3 + 16x^2 - 10x$ |
| 6. $6x + 5$ | 23. $13x - 9y + 2$ | 40. $18x^3 - 12x^2 + 22x$ |
| 7. $15x - 6y - 10$ | 24. $21x + 29y + 4$ | 41. $26x^3 - 4x^2 - 2x$ |
| 8. $13x + 21y + 2$ | 25. $13x + 18y - 5$ | 42. $7x^3 + 9x^2 - 4x$ |
| 9. $x + 9y - 4$ | 26. $-8x^2 + 11x$ | 43. $27x^3 + 37x^2 - 25x$ |
| 10. $-3x + 10y + 7$ | 27. $-6x^2 + 7x$ | 44. $17x^3 + 17x^2 + 31x$ |
| 11. $11x + 18y + 3$ | 28. $-15x^2 + 5x$ | 45. $19x^3 + 29x^2 - 12x$ |
| 12. $5x - y - 5$ | 29. $-6x^2 + 9x$ | 46. $2x^3 - 7x^2 + 2x$ |
| 13. $2x - 6y + 8$ | 30. $-7x^2 + 15x$ | 47. $-23x^3 + 7x^2 + 10x$ |
| 14. $13x + 20y + 1$ | 31. $23x^2 - 5x$ | 48. $-6x^3 + 11x^2 + 33x$ |
| 15. $21x + 9y + 3$ | 32. $19x^2 + 6x$ | 49. $16x^3 + 24x^2 + 15x$ |
| 16. $-14x + 10y + 5$ | 33. $17x^2 + 11x$ | 50. $-7x^3 - 10x^2 - 9x$ |
| 17. $x + 10$ | 34. $9x^2 + 10x$ | |

Part D: Properties of Addition and Multiplication

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|--|--|---|
| 1. A) Commutative Property of Addition | 11. C) Associative Property of Addition | 21. D) Inverse Property of Addition |
| 2. C) Commutative Property of Multiplication | 12. C) Associative Property of Addition | 22. C) Inverse Property of Multiplication |
| 3. A) Commutative Property of Addition | 13. D) Identity Property of Multiplication | 23. B) Inverse Property of Multiplication |
| 4. C) Commutative Property of Addition | 14. C) Identity Property of Multiplication | 24. C) Inverse Property of Multiplication |
| 5. D) Commutative Property of Multiplication | 15. B) Identity Property of Addition | 25. C) Commutative Property of Addition |
| 6. A) Commutative Property of Addition | 16. D) Identity Property of Addition | 26. C) Associative Property of Addition |
| 7. D) Associative Property of Addition | 17. A) Identity Property of Addition | 27. B) Commutative Property of Multiplication |
| 8. B) Associative Property of Addition | 18. B) Identity Property of Addition | 28. D) Identity Property of Multiplication |
| 9. C) Associative Property of Multiplication | 19. A) Inverse Property of Multiplication | 29. A) Inverse Property of Multiplication |
| 10. D) Associative Property of Addition | 20. A) Inverse Property of Addition | 30. A) Associative Property of Multiplication |

Part E: Solving Equations

- | | | |
|-------------------------|------------------------|--------------|
| 1. $x = -6$ | 18. $x = 8$ | 35. $x = -7$ |
| 2. $x = 2$ | 19. $x = 7$ | 36. $x = 2$ |
| 3. $x = 6$ | 20. $x = -\frac{8}{3}$ | 37. $x = -4$ |
| 4. $x = 4$ | 21. $x = 2$ | 38. $x = 12$ |
| 5. $x = 8$ | 22. $x = 4$ | 39. $x = 8$ |
| 6. $x = -2$ | 23. $x = 3$ | 40. $x = -2$ |
| 7. $x = 4$ | 24. $x = 3$ | 41. $x = 4$ |
| 8. $x = 3$ | 25. $x = -12$ | 42. $x = -1$ |
| 9. $x = 4$ | 26. $x = -11$ | 43. $x = 2$ |
| 10. $x = -8$ | 27. $x = -\frac{1}{4}$ | 44. $x = 6$ |
| 11. $x = 6$ | 28. $x = \frac{1}{3}$ | 45. $x = -5$ |
| 12. $x = 6$ | 29. $x = 8$ | 46. $x = -7$ |
| 13. $x = -1$ | 30. $x = 7$ | 47. $x = 8$ |
| 14. $x = -8$ | 31. $x = 2$ | 48. $x = -9$ |
| 15. $x = 7$ | 32. $x = -4$ | 49. $x = 7$ |
| 16. $x = -\frac{11}{3}$ | 33. $x = -4$ | 50. $x = 48$ |
| 17. $x = \frac{11}{4}$ | 34. $x = 1$ | |